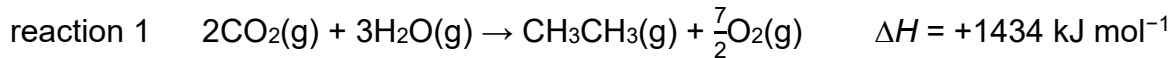


9 Consider the following reactions.



The standard enthalpy change of combustion of ethane is $-1542 \text{ kJ mol}^{-1}$.

Which statements are correct?

1 y is +36

2 At 373 K, ΔS for reaction 2 is $\frac{y}{373} \text{ kJ mol}^{-1} \text{ K}^{-1}$.

3 Reaction 1 is spontaneous only at high temperatures.

A 1 and 2 only **B** 1 and 3 only **C** 2 and 3 only **D** 2 only